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The concept of using of neural network computer systems to assist establish Radiocommunications in Distress situations for captains of tourist yachts sailing around the Lesser Antilles.

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Abstract

The Lesser Antilles Sea Area is a particularly dangerous area for small unconventional, including, tourist sea yachts because the range of available communications in the areas of communication zones A1 and A2, is severely limited there. This represents a significant constraint on the safety of navigation for these vessels because in a dangerous situation requiring a call for RCC assistance, the use of common VHF/MF radio equipment for this purpose is limited and often ineffective. The ability to contact Maritime Rescue Coordination Centers in case of danger to life and property on a yacht is an important issue. Recently, there has been an increased number of serious maritime accidents involving cruising sea yachts, during which there have been serious problems in establishing radio communication in dangerous situations. The article presents the concept of an intelligent radio communication system to support the decision of the captain of a small sea tourist yacht, sailing in the waters near Lesser Antilles, in a dangerous situation requiring a call for RCC assistance. The discussed new system, based on the method of artificial neural networks ANN, makes it possible to help the captain make the right navigation decision in an emergency situation. The content presents the calculation model and an example of numerical simulation of the operation of the described decision support system, with reference to the existing communication areas A1, A2 Lesser Antilles area and assuming a given position of a tourist small yacht in a dangerous situation, requiring a call for RCC assistance by means of VHF/MF radio equipment. The result of the simulation is the calculation of the distance to the nearest area to establish communication with the Maritime Rescue Coordination Center and the value of the new corrected course of the tourist yacht and the estimation of the time required to reach the aforementioned area.

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1. Introduction

The cruising area around the archipelago of the Lesser Antilles is considered one of the most dangerous areas for sailing in the world. In this area every year there are many marine accidents, especially small tourist yachts. One example, among others, is the disaster of the sailing yacht type Bavaria 44 m/y "Vagant" on November 11, 2017, during which, while sailing from Europe to Barbados, the 74-year-old captain fell overboard and his wife had trouble steering the yacht, taking proper rescue action and calling for help via radio communication equipment. As a result, it was too late for the rescue, the captain drowned in the middle of the Atlantic, and the zona two days later went aboard a merchant ship passing nearby and the abandoned yacht, after drifting in the ocean for several weeks, was dumped on Lambeau Reef near Tobago in the Lesser Antilles archipelago (see Fig.1). Note that the area is bordered to the east by the open Atlantic Ocean and to the north by the famous Bermuda Triangle, famous for unexplained disasters and other maritime events. Many tourist yachts coming from the US, Europe and the Caribbean sail in the area. Unfortunately, despite its attractiveness and popularity as a tourist destination, the individual islands that make up the Lesser Antilles archipelago are characterized by poor radio communications infrastructure, especially with regard to Class A1 and A2 communication areas for VHF and medium MF radio communications. Therefore, according to [2], the described region is a white spot in the VHF/MF radio security system in the world. The nearest official radiocommunication stations in the area are Curacao and San Juan leaving the archipelago of the Lesser Antilles without radiocommunication cover (see Fig.2.). Unfortunately, most small cruising yachts and non-conventional vessels under SOLAS regulations are not required to have advanced radiocommunications equipment to establish communications on higher HF frequencies in the A3 area. This is also due to the high cost of such equipment compared to common VHF/MF radios. This creates a particularly dangerous situation for those sailing in the area, who do not have HF radios on board in case of a dangerous situation and the need to urgently call for help. Taking into account also the fact that many captains of small yachts are not adequately trained how to behave in dangerous situations, such



Fig. 1 The route and location of the catastrophe of the sailing yacht type Bavaria 44 m/y "Vagant", which was involved in a maritime accident on November 21, 2017 while sailing from Europe to Barbados in the Lesser Antilles.

Source: Source: materials from the official post-accident report of the Polish National Commission for the Investigation of Marine Accidents [3], published by PKBWM Poland,

a condition poses a potential danger to the safety of the yacht and the lives of the crew and passengers on board. The subject of this article is to develop a concept of a decision support system for yacht captains to calculate the corrected course and the shortest route to an area where it will be possible to establish VHF/MF radiocommunications and effectively call for help from the RCC/SAR Maritime Rescue Coordination Center. To this end, the authors of the article proposed a new decision support system for the yacht's captain, which works using artificial intelligence tools to help the confused captain make the right decision and calculate the distance to the nearest area, set a new corrected course of the vessel to estimate the required time to reach the above area and call for help.

2. Situation with Radiocommunication areas around the Lesser Antilles archipelago designated by the administrations of countries

The areas of sea waters around the archipelago of the Lesser Antilles are particularly dangerous in terms of RCC radio surveillance for the navigation of yachts and smaller non-conventional vessels. This is illustrated in Fig. 3. In the described area there are, according to the provisions of the International Convention for the Safety of Life at Sea - SOLAS, three areas of RCC-SAR rescue responsibility, defined as follows [5,11,16,19]:

- A1 - maritime area covered by at least one VHF DSC shore station in which continuous emergency communication is provided and which is established by the administration of the country,
- A2 - maritime area covered by at least one MF DSC shore station (outside the A1 area) in which continuous emergency communication is provided and which is established by the administration of the country concerned,
- A3 - maritime area covered by Inmarsat geostationary satellites (outside areas A1 and A2), where continuous emergency communications are provided,

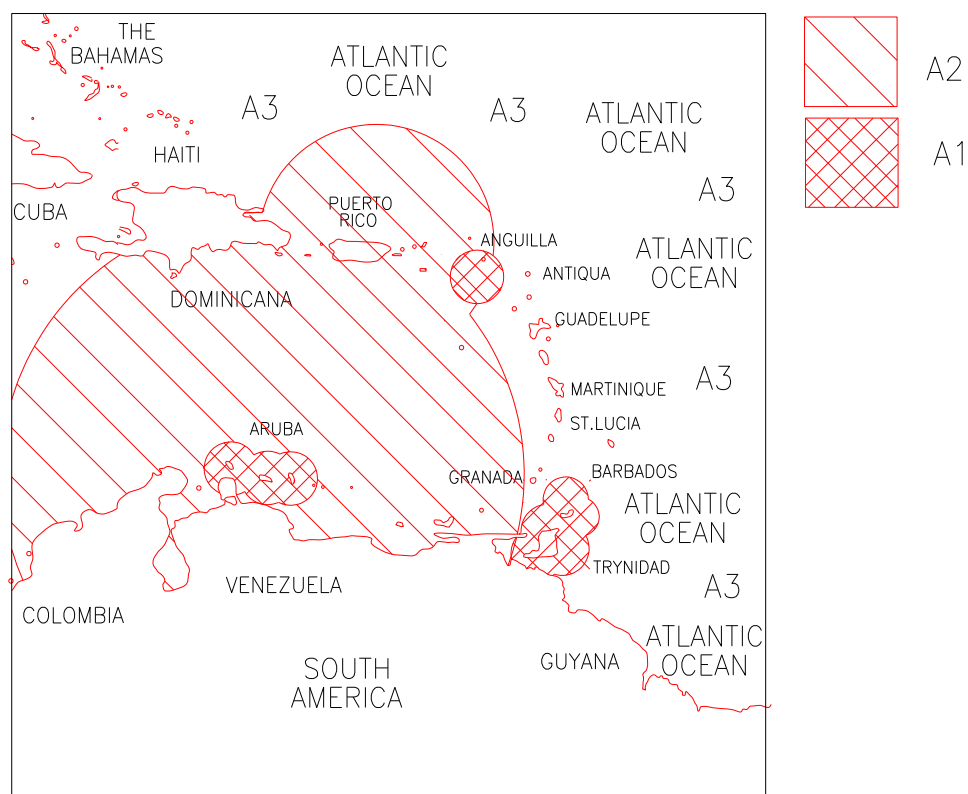


Fig. 2 Lesser Antilles Area division into Zones of Responsibility for Rescue Operation by RCC service

Source: Own preparation based on ALRS (Admiralty List of Radio Signals) Admirals Chart 2021/22, Vol 5, pp.165, published by UK, Hydrographic Office, [5]

As can be seen in Fig. 2, from the east of the archipelago and also around most of the archipelago extends only the A3 area requiring possession of advanced radiocommunications equipment transmitting using HF long wave. Under SOLAS regulations, most small cruising yachts and non-conventional vessels are not required to have such advanced radiocommunications equipment, especially since it is expensive and requires special skills to operate. Only in the northern and southern parts of the archipelago do the waters fall under the A1 radiocommunication surveillance area, which allows the establishment of communications on VHF. Unfortunately range A1 is limited to a distance of several tens of NM from VHF transmitting stations. From the west, at a considerable distance extends the A2 area served by a shore station operating at mid-frequency waves of the MF DSC system. Such a situation is particularly dangerous. It means that in the area most popular in yacht tourism, that is, east of the archipelago of the Lesser Antilles is unfortunately dominated by the zone covered by Inmarsat A3 geostationary satellites. This means that the majority of tourist yachts that are only equipped with DSC VHF or even DSC MF radios will not have the ability and sufficient range to establish communications with the RCC/SAR rescue center and effectively call for SAR assistance in the event of a dangerous situation on board. So, this creates a great danger for the crews of cruising yachts and other smaller non-conventional vessels in the described area in case of a dangerous situation on board of them for life and property.

3. Problem formulation

Since the main problem of navigational hazards in the Lesser Antilles archipelago concerns smaller pleasure yachts and other smaller non-conventional vessels, i.e. ships of less than 300 GT on which there is no requirement under the SOLAS Convention to have advanced and expensive HF DSC radiocommunication equipment, let's assume in the case under consideration that our yacht is equipped with a DSC VHF/DSC radio station (providing communications in areas A1 and A2). Let's assume at the same time that the crew is certified and has the skills to operate them correctly. The case in question is a typical situation in which the described tourist yacht sails at a considerable distance from the island of the archipelago from the eastern side. It should be recalled here that the vast majority of yachts sailing from Europe to the Caribbean (and on the return voyage) are at the end of crossing the Atlantic Ocean in such a situation.

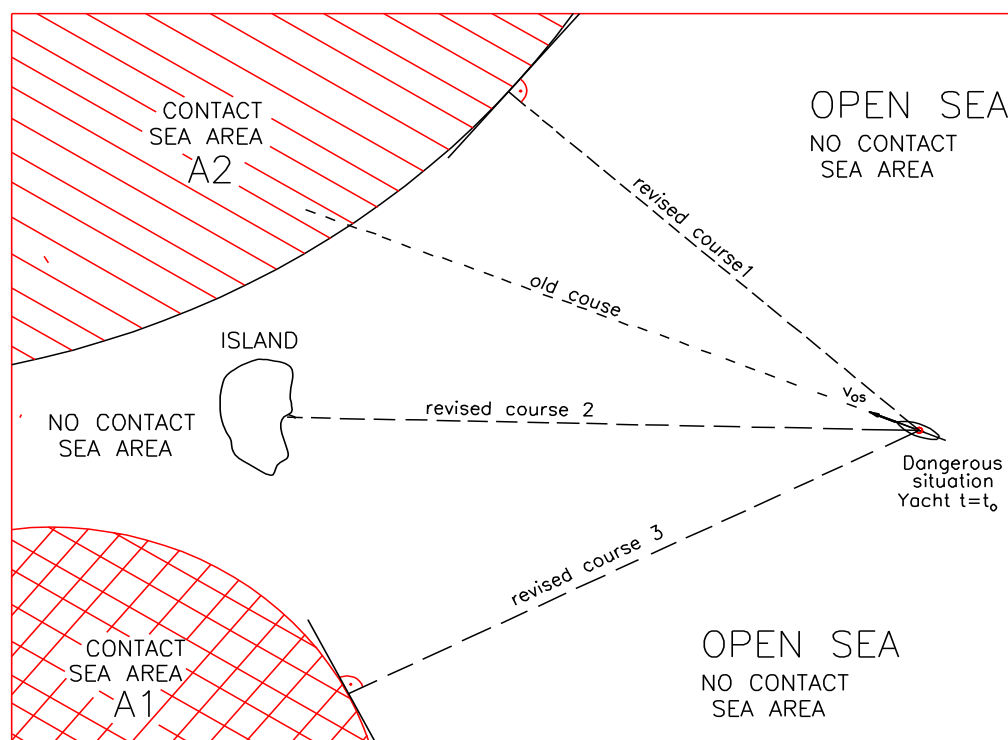


Fig. 3 Fig. 3 Illustration of the formulated research problem

Source: Own preparation

At the time $t=t_0$, a dangerous situation occurred on board the yacht - related to danger of life - for example, a man falling overboard or a collision of the yacht with another vessel (yacht) on the high seas). At the moment under consideration, the yacht is away from the island in the HF/A3 communications area (see the situation illustrated in Fig. 3). Since the analyzed yacht is only equipped with standard radio communication equipment, RTF/VHF/MF, establishing direct communication with the nearest RCC Rescue Coordination Center by the captain to call for help and initiate rescue operations is not possible due to the lack of active radio coverage. This creates a potentially dangerous situation requiring the skipper to make the right navigational decision. The MF/ A2 medium wave communication range is located to the north of the island, while the A1 area for establishing VHF communication is located to the south. . the yacht's captain therefore has to decide whether the yacht should sail by the shortest route - as quickly as possible - to the A1 communication area or to the A2 area, or finally sailing to the island in order to be able to establish effective radiocommunication and call for help from the RCC/SAR maritime rescue center. The decisive criterion for choosing the optimal navigational decision, therefore, will be the time criterion , when choosing which option will be possible to effectively call for help as soon as possible .

Let's assume that at the time of the occurrence of a dangerous situation, the analyzed yacht is in a position defined by the following position coordinates:

$$A_0(t=t_0) = \{ X_{t_0}; Y_{t_0} \} \quad (1)$$

where : X_{t_0}, Y_{t_0} -coordinate x,y of the yacht at the time of the emergency situation $t = t_0$

The mathematical notation of contact A2 area limit line is the set of points as follows:

$$CLLA_2 = CLLA_2(XA_{2t}; YA_{2t}) = \{ XA_{2ti}; YA_{2ti} \} \quad (2)$$

where : $XA_{2ti}; YA_{2ti}$ - coordinates of the points belonging to the contact limit line A2 area

The distance, from the location of the dangerous situation on the yacht A_0 to the Harbor on the nearest island IS ` can be calculated from the following equation :

$$|A_0IS| = \sqrt{(X_{t_0} - X_{IS})^2 + (Y_{t_0} - Y_{IS})^2} \quad (3)$$

where : $X_{IS}; Y_{IS}$ - coordinates of the Harbor on the nearest island IS

Similar to the notation of contact A2 area limit line, the mathematical notation of contact A1 area limit line is following:

$$CLLA_1 = CLLA_1(XA_{1t}; YA_{1t}) = \{ XA_{1tj}; YA_{1tj} \} \quad (4)$$

where : $XA_{1tj}; YA_{1tj}$ - coordinates of the points belonging to the contact limit line A1 area

The distance that a yacht must travel from the place where the danger occurred to the border of the A2 communication zone ` can be calculated from the formula :

$$\overline{A_0CLLA_2}_{tti} = \sqrt{(X_{t_0} - XA_{2ti})^2 + (Y_{t_0} - YA_{2ti})^2} \quad (5)$$

The same calculation referring go to Yacht travel from the place where the danger occurred to the border of the A2 communication zone ` can be calculated from the relation :

$$\overline{A_0CLLA_1}_{ttj} = \sqrt{(X_{t_0} - XA_{1tj})^2 + (Y_{t_0} - YA_{1tj})^2} \quad (6)$$

The mathematical problem to solving and calculation following values :

1. Closest distance of Approach the active A2 radiocommunication zone

$$CDAA2 = CDAA2(x_{to}, y_{to}, \{x_{ti}, y_{ti}\}) \quad (7)$$

$$CDAA2 = \text{opt } \overline{A_o C L L A 2}_{t t i} \rightarrow \min (\overline{A_o C L L A 2}_{t t i}) \text{ for } \mathbf{C L L A 2 t} = \mathbf{C L L A 2 t}(X_t; Y_t) = \{X_{t i}; Y_{t i}\} \quad (8)$$

2. Time of Closest distance of Approach the active A2/MF connection Area

$$TCDA A 2 = TCDA A 2(CDAA2, v_{os},)$$

where: TCDA A 2 – time in which the yacht, sailing at a given speed AND with a corrected course , will reach contact area A2.

$$TCDA A 2 = \frac{CDAA2 = \min (\overline{A_o C L L A 2}_{t t i})}{v_{os}} \quad (9)$$

3. Time of Closest distance of Approach the harbor on nearest Island

$$TCDA IS = \frac{|A o IS|}{v_{os}} \quad (10)$$

4. Closest distance of Approach the active A1 radiocommunication zone

$$CDAA1 = CDAA1(x_{to}, y_{to}, \{x_{tj}, y_{tj}\}) \quad (11)$$

$$CDAA1 = \text{opt } \overline{A_o C L L A 1}_{t j} \rightarrow \min (\overline{A_o C L L A 1}_{t j}) \text{ for } \mathbf{C L L A 1 t} = \mathbf{C L L A 1 t}(X_t; Y_t) = \{X_{t j}; Y_{t j}\} \quad (12)$$

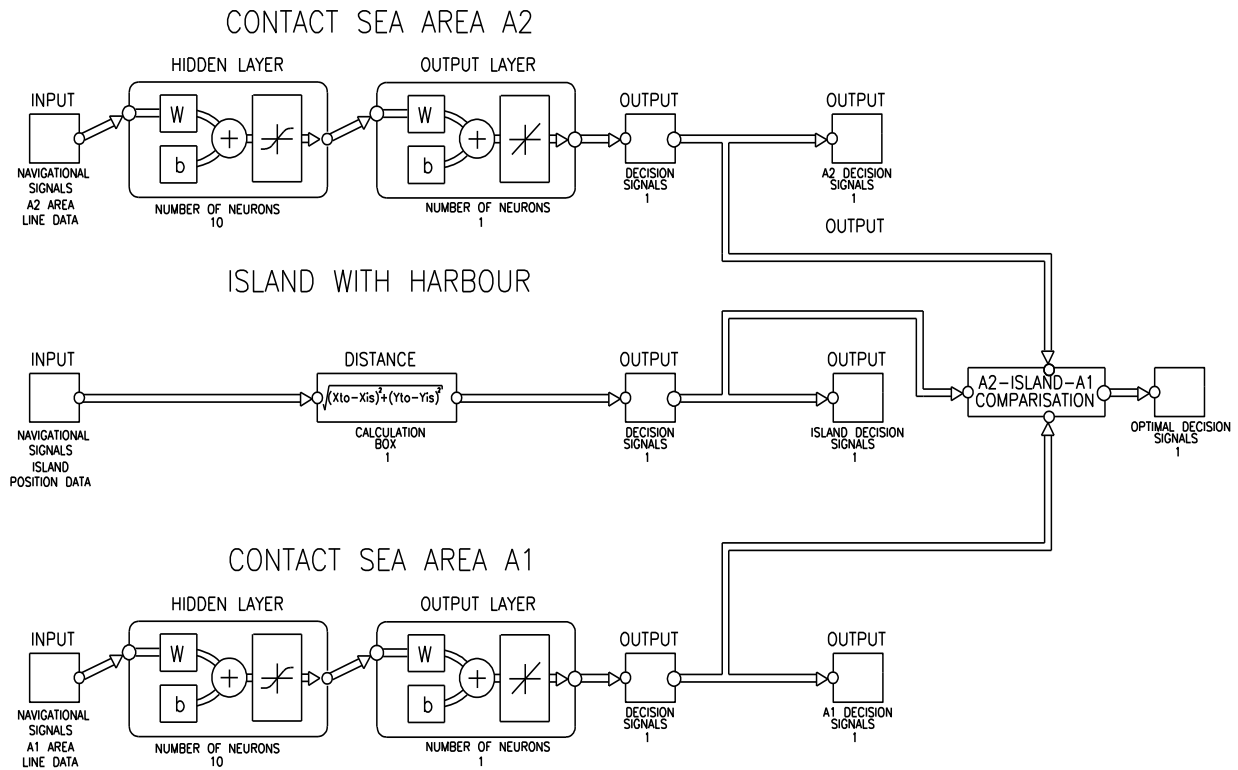


Fig. 4 Structure of Artificial Neural Network System where: n=6 - number of decision variables (simplified example)

Source: Own preparation based on Matlab r2020b ffp. App. [10]

5. Time of Closest distance of Approach the active A1/VHF connection Area

$$TCDA1 = TCDA1(CDAA1, v_{os},)$$

Solving the problem same as presented in point 2, we can shown

$$TCDA1 = \frac{CDAA1 = \min(\overline{A_0 C L L A 1}_{tj})}{v_{os}} \quad (13)$$

Finally, from the comparison of the two times $\{TCDA2, TCDA2\}$, the optimum decision time $TCDA_{opt}$ can be determined as the shortest time in which a yacht, sailing at a given speed, will reach any active contact zone for connecting with the RCC rescue station and sending a distress signal to the SAR service.

$$TCDA_{opt} = \min\{TCDA1, TCDAIS, TCDA2\} \quad (14)$$

4. Use of the Artificial Neural Network System for calculation CDA_{opt} and $TCDA_{opt}$ with revised Yacht Course

The problem formulated in the previous chapter has a strongly nonlinear character. To develop a methodology for its solution, it was decided to use artificial neural network algorithms ANN [4,8,15]. Their use will allow a quick and

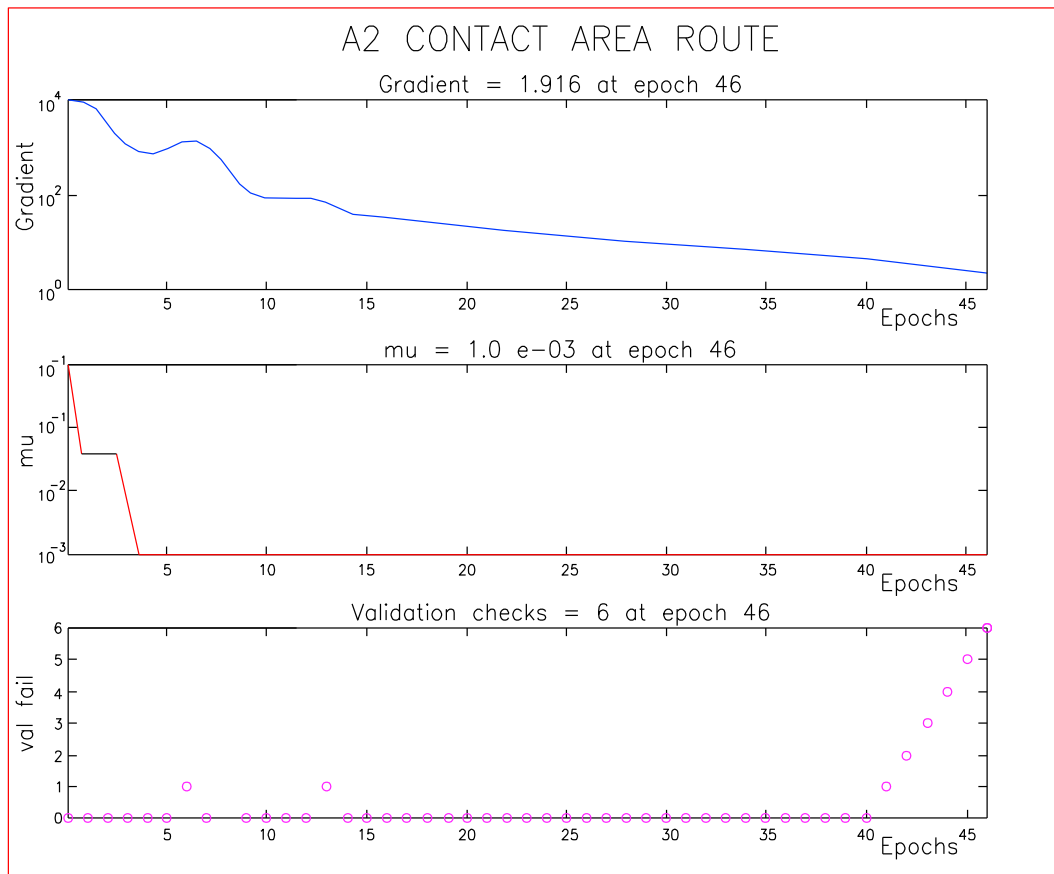


Fig.5. A1 Route and A2 Route ANN system Learning process

Source: Own preparation based on results from the computer calculation ANN system Matlab ver.R2020b [16]

simple solution, based on the implementation of the provided initial data as input. The simplicity and speed of calculations here is a practical advantage of the concept described, as part of a decision support system for the captain of a yacht or a small non-conventional vessel. This constitutes a great practical advantage of the above solution. Fig.4 shows the structure of the ANN-Artificial Neural Network system, which was used in the simulation of the optimization process. It consists of two parallel ANN routes with the first one being used to calculate the Closest distance of Approach the active A2 radiocommunication zone CDAA2 and the time value TCDA2 where are the set of coordinates of the contact boundary line **CLLA2** for the coastal section of the analyzed sea area was input:

$$\text{CLLA2}(\text{XA2t}; \text{YA2t}) = \{ \text{XA2ti}; \text{YA2ti} \}$$

and the position of the yacht in distress its course and current speed :

$$\text{Aos} = \text{Aos}(\text{Xos}; \text{Yos}); \quad \text{course } \varphi_{\text{os}}; \quad \text{yacht's speed } v_{\text{os}}.$$

Analogous to the previous one, the second route ANN , is used to calculate the Closest distance of Approach the active A2 radiocommunication zone CDAA1 and the time value TCDA1 where a set of **CLLA1** contact boundary line coordinates and yacht data were entered as an input data set:

$$\text{CLLA1}(\text{XA1t}; \text{YA1t}) = \{ \text{XA1ti}; \text{YA1ti} \}$$

Decision signals from both ANN routes were compared with each other to determine the optimal yacht route and optimal TCDAopt time. The ANN system presented was used to simulate the critical situation of a yacht out of communication range of A2, IS and A1. And the developed results are a function of the objective - to develop decision support for the captain on what corrected course the yacht must follow in order to be in the A2, IS or A1 communication zone as soon as possible, which would enable an effective call for help by RCC-SAR [6, 12].

5. Examples of calculation results

For the solution of the optimization task presented in the previous chapter, based on finding the optimal path for the yacht to reach the area from which it will be possible to establish communication with the RCC Maritime Rescue Coordination Center station using a VHF or MF radio, a numerical simulation was carried out using the neural network model shown in Fig.3 . For the calculations, the computer program MATLAB ver R 2020b update 4 (9.9.0.15700001) 64-bit (win64) from Mathworks Inc. Company [10]. The yacht at time $t=t_0$, at the time of the life-threatening event on the vessel, was located at a point with coordinates $\text{Aos}(\text{Xos}; \text{Yos})$ (see situation sketch in Fig.3)

That is, outside the active areas A2, IS and A1. It was sailing at the time of the incident at a speed of $v_{\text{os}}=12.00$ kn (6.17 m/s) with the yacht's initial course $\varphi_{\text{os}}=292$ deg NW directed towards the open sea.

The boundary line of the A1 active zone was mapped using a set of 140 points [3, 4, 13,15]:

$$\text{CLLA2} = \text{CLLA1}(\text{XA2t}; \text{YA2t}) = \{ \text{XA2ti}; \text{YA2ti} \}, \quad \text{where } i=1,2,\dots,140$$

Localization of Harbor on nearest island is described as following coordinates:

$$\text{IScoordinates} = (\text{XIS}; \text{YIS})$$

In contrast, the boundary line of the A2 active zone was mapped using a set of 115 points :

$$\text{CLLA2} = \text{CLLA2}(\text{XA2t}; \text{YA2t}) = \{ \text{XA2tj}; \text{YA2tj} \}, \quad \text{where } j=1,2,\dots,115$$

In simulation calculations , according to the assumptions, did not take into account wave, tidal or wind force and its direction. Presented The proposed model m ANN Matlab environment solved the optimization tasks:

$$\begin{aligned} \text{CDAA2} &= \text{opt } \overline{A_o \text{CLLA2}_{tti}} \rightarrow \min (\overline{A_o \text{CLLA2}_{tti}}) \text{ for } \text{CLLA2t} = \{ \text{Xti}; \text{Yti} \} & (\text{Route ANN A2}) \\ |A_o \text{IS}| \sqrt{(\text{Xto} - \text{XIS})^2 + (\text{Yto} - \text{YIS})^2} & & (\text{Route ANN IS}) \\ \text{CDAA1} &= \text{opt } \overline{A_o \text{CLLA1}_{tj}} \rightarrow \min (\overline{A_o \text{CLLA1}_{tj}}) \text{ for } \text{CLLA1t} = \{ \text{Xtj}; \text{Ytj} \} & (\text{Route ANN A1}) \end{aligned}$$

The course of the learning process of the detailed routes A2 AND A1 is presented in Fig.5, with the former case of A2 ending after 46 epochs and the latter after 37 epochs.

As a result of the calculations, the following calculation result was obtained:

{CDAopt = CDAA2 = 168,77 NM with the corrected new course of the yacht $\phi_{\text{osr}}=312$ deg NW}

The time to reach zone A2, enabling contact with the RCC rescue center via MF radio/radiotelephony, was : TCPAopt = TCPAA2=14 h 4 min, assuming unchanged yacht speed $v_{\text{os}}=12.00$ kn (6.17 m/s). In the case of the option of reaching the A2 zone correspondingly obtained :

The time to reach the harbor on island is TCPAIS =16 h 3 min,
{|AoIS| = 192.69 NM with corrected course of the yacht $\phi_{\text{osr}}=271$ deg NW}.

The time to reach the A1 zone, enabling contact with the RCC rescue center by means of VHF radio/radiotelephony, was : TCPAA1 =16 h 14 min,

{CDAA1 = 194.87 NM with corrected new course of the yacht $\phi_{\text{osr}}=243$ deg SW}.

The results of the presented simulation are an important element of decision support for the yacht captain in an emergency situation. Based on them, the yacht's helmsman can make the right navigational decision in an emergency situation. This can be of great practical importance for improving the safety of life at sea for yachts navigating in the area of the Lesser Antilles archipelago.

6. Conclusions

The region of the sea waters around the archipelago of the Lesser Antilles is classified as one of the most dangerous sailing areas for smaller cruising yachts and other non-conventional vessels. A large part of the archipelago is not covered by radio communication area types A1 and A2, which makes it impossible to establish radio communication in an emergency situation using common VHF/MF radio means. This causes a particularly dangerous situation that threatens the safety of tourist navigation in the area. One of the solutions that can improve the safety of navigation is the decision support system proposed by the authors of the article for the captain of the yacht in a dangerous situation. It allows, with the use of numerical artificial neural network technology, to analyze the current situation of the yacht and determine the corrected most favorable course in order to sail as quickly as possible to the radio communication area or island to effectively establish radio communication with the nearest Rescue Coordination Center RCC and call for SAR assistance.

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